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Erasmus+ project ROBO STEAM ACADEMY
KA220-VET-7CF4F308

Lap Timer Box Assembly Guide

Print, inspect and assemble the version 2 fit prototype · Version 2.0

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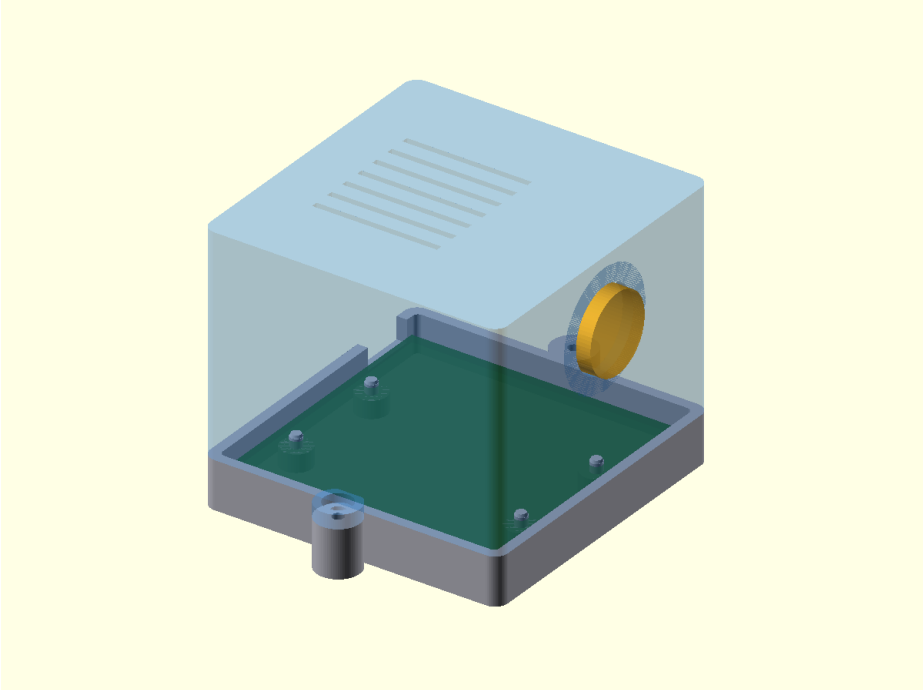
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About this guide

This guide shows how to print and mechanically assemble the version 2 lap-timer box. It is written for students aged 16 to 19, teachers, and competition organisers. Version 2 corrects the board location and raises the sensor after the first physical test. It remains a fit-test prototype. Record every part that does not fit before changing the editable OpenSCAD model.



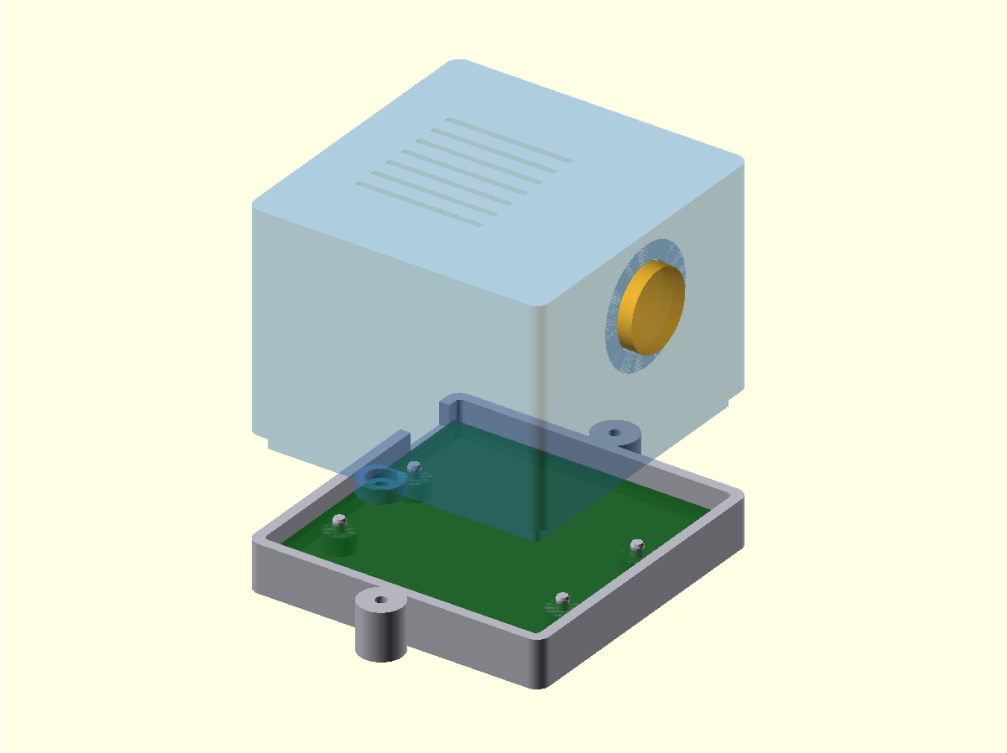
Version 2 enclosure with the sensor and board envelopes shown for inspection.

> **Safety**
>
> A 3D-printer nozzle and build plate can cause burns. Wait until the printer reports a safe temperature before removing a part.

1. Understand the enclosure

The enclosure has two printed main parts. The bottom tray holds the HW-389-style baseboard and NodeMCU. The removable top holds the E18-D80NK sensor. Two external side lugs keep the screws outside the PCB area.

Design item	Version 2 value	Status
Sensor arrangement	Centred	Selected Design A
Sensor optical-axis height	45 mm	User-selected CAD nominal
Main enclosure height	61 mm	CAD nominal
Main shell footprint	69.1 x 70.3 mm	CAD nominal, excluding lugs
Sensor panel opening	19.0 mm	Passed version 1 physical fit test
Nominal walls and floor	2.4 mm	Starting value



Exploded view of the version 2 base, board envelope, sensor and removable top.

1.1 Why this is a prototype

The PCB outline and mounting centres come from the user's complete 600 dpi scan. The model gives 1.0 mm clearance at the rear connector edge and 3.0 mm at the left, right, and front edges. The rear connector positions, sensor length, exact board-hole diameter, assembled electronics height, and screw engagement still require a physical check.

> **Note**
>
> Do not force a board, plug, sensor, or screw. A poor fit is useful version 2 test data. Measure it and revise the model.

2. Prepare the files

1. Download the five version 2 STL files from `3d_printing_files`.
2. Open each STL in the slicer.
3. Keep the base with its floor on the build plate.
4. Keep the top in its supplied roof-down orientation.
5. Slice the PCB fit coupon before the main parts.
6. Slice the small sensor and screw tests if those interfaces still need checking.

Check: the slicer shows a base, a top, a PCB fit coupon, a flat sensor-hole plate, and a rectangular screw coupon.

2.1 Starting print settings

Use a 0.4 mm nozzle and 0.2 mm layer height as the starting point. PLA or PETG may be used. These settings have not yet been confirmed by a successful prototype print. Save the actual slicer profile used.

> **Tip**
>
> Print the PCB fit coupon first. It checks the changed board outline and all four locating pegs with much less material than the complete enclosure.

3. Test the sensor opening

1. Print `sensor_nut_check_v2.stl` if a second printer or material must be checked.
2. Let the part cool.
3. Remove any brim without enlarging the hole.
4. Insert the threaded M18 sensor body gently.
5. Check that the retaining nut can clamp the plate.
6. Measure the printed hole with calipers.
7. Write the result in `OPEN_QUESTIONS_v2.md`.

Check: the sensor enters without cracking the plate and does not wobble after the nut is tightened by hand.

> **Safety**
>
> Do not look closely into an energised infrared sensor. The E18-D80NK is treated as an infrared source; project evidence does not identify it as a laser.

4. Test the closure screw

1. Print `screw_coupon_v2.stl` in the same material and orientation as the base.
2. Let the coupon cool.
3. Try the intended 3 mm screw in each test hole.
4. Stop if the plastic begins to split.
5. Select the hole that gives firm engagement without visible cracking.
6. Record the screw type and result.

Check: the chosen screw can be inserted and removed without splitting the coupon.

5. Test the PCB outline and locating pegs

1. Print `pcb_fit_coupon_v2.stl` with its flat face on the build plate.
2. Let the coupon cool before removing it.
3. Place the bare HW-389-style baseboard above the four tapered pegs.
4. Confirm that every peg enters its hole without bending the PCB.
5. Confirm that the board does not touch the left, right, or front coupon walls.
6. Confirm that the 1.0 mm rear allowance places the connector edge correctly.

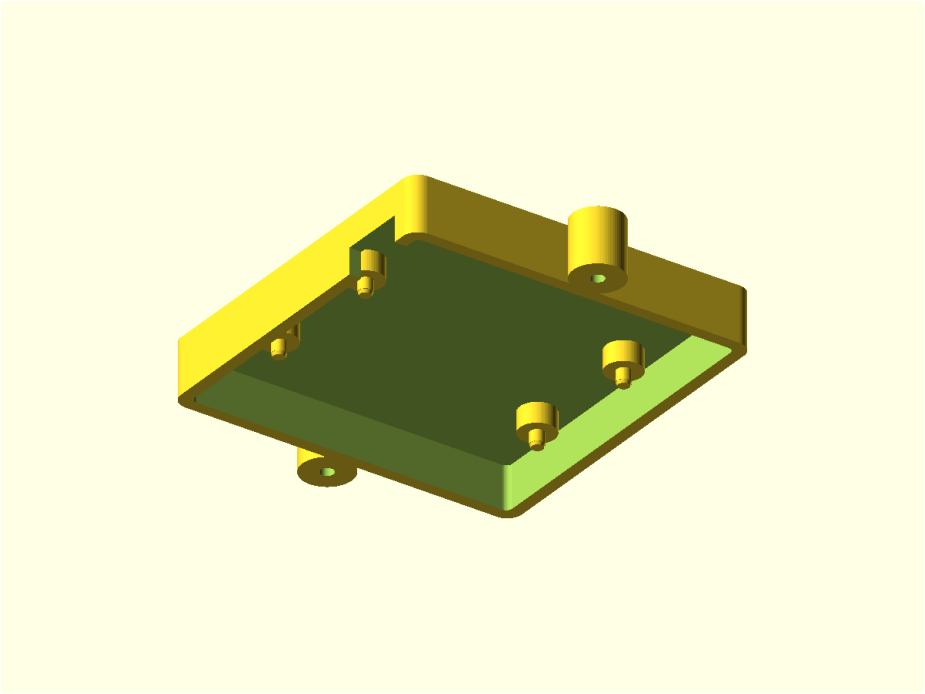
Check: the board rests on all four standoffs, lifts off without force, and is not clamped by the coupon walls.

6. Print and inspect the main parts

1. Print `base_v2.stl`.
2. Print `top_v2.stl`.
3. Let both parts cool before removal.
4. Remove strings and brim material without cutting functional surfaces.
5. Check that the locating lip enters the base without force.

6. Check that both external lug holes line up.
7. Check that the rear openings are clear.

Check: the empty top sits fully on the base and the seam is even on all four sides.



Version 2 bottom tray showing the four tapered locating pegs.

7. Fit the electronics without power

1. Disconnect every power source.
2. Place the HW-389-style baseboard over the four locating pins.
3. Stop if any pin bends the PCB or misses a hole.
4. Confirm that the connector edge faces the rear openings.
5. Install the NodeMCU on the baseboard in its correct hardware orientation.
6. Place the top on the base without fitting the sensor.
7. Insert the largest micro-USB plug that must be supported.
8. Insert the intended DC barrel plug.
9. Confirm that each plug seats fully and can be removed by its body.

Check: the board is not stressed and both plugs can be fully inserted with the box closed.

> **Safety**
>
> Do not connect a mains adapter until its output voltage, polarity, current rating, and suitability for the project have been approved.

8. Fit the sensor

1. Remove the top.
2. Pass the E18-D80NK body through the front opening from outside.
3. Leave the sensing face and front retaining nut outside.
4. Fit the inner nut without pinching the wall.
5. Route the cable in a gentle curve.

6. Leave a service loop so the top can be lifted away.
7. Place the top on the base.
8. Look through the open seam and check clearance above the NodeMCU and wiring.
9. Measure the smallest clearance.

Check: the sensor body and cable do not touch the NodeMCU, connectors, or wiring.

9. Close the enclosure

1. Disconnect power.
2. Lower the top while watching the service loop.
3. Confirm that no wire crosses the locating lip.
4. Insert the two selected screws from above.
5. Tighten each screw only until the seam closes.
6. Stop if a lug bends or a boss cracks.

Check: the top is secure, the seam is even, and the sensor beam remains horizontal.

10. Electrical completion and first test

The project repository does not yet contain an authoritative wiring diagram, firmware source, pin map, approved adapter specification, or verified first-test indications. Complete those steps only from the approved project sources when they are added.

> **Safety**
>
> Never guess a sensor pin or power polarity. A wrong connection can damage the NodeMCU, baseboard, sensor, or computer.

Check: the mechanical prototype is labelled with its version and all fit results are recorded before any electrical test.

11. Record the version 2 result

Photograph the assembled box from the front, rear, top, and side. Record the printer settings, hole fits, connector fits, sensor clearance, screw result, and every CAD change required for version 3.

> **Challenge**
>
> Compare the nominal drawing dimensions with the printed part. Explain which differences come from slicing, material shrinkage, or measurement technique.

About this document



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